IMPLEN

inkarp®
Your Knowledge... Our Solution...!

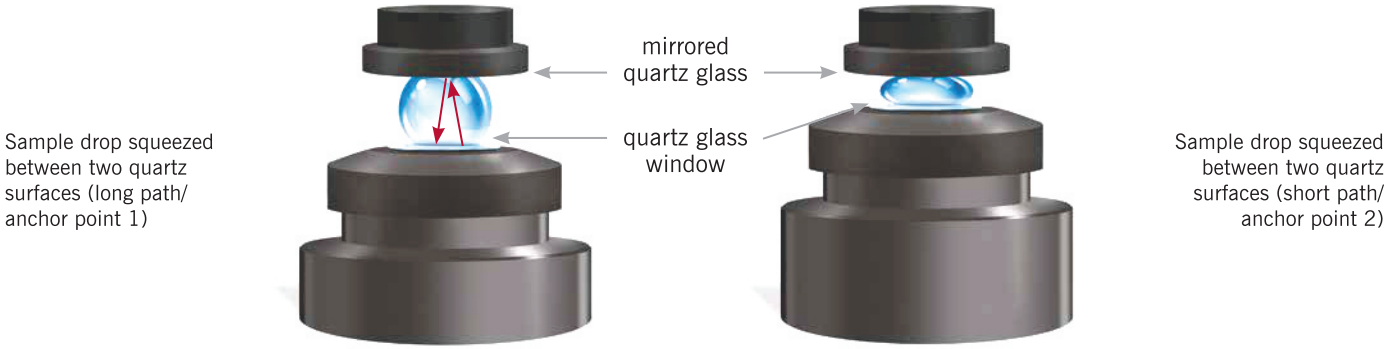
NanoPhotometer®

NanoVolume and Cuvette Spectroscopy



Sample Compression Technology Accuracy & Precision

The NanoPhotometer® uses Sample Compression Technology acting like a microscope slide and a coverslip to form a capillary film between two quartz surfaces with each sample, eliminating the need for surface tension. This allows for accurate readings of proteins and the ability to use organic solvents as well as detergent buffers. The NanoPhotometer® works with two precisely defined fixed path lengths to guarantee linearity over the full detection range of the instrument.



NanoPhotometer® Family Features

Sample Compression Technology
Uniquely contained sample environment avoids evaporation and the need for surface tension.

Vortex
Ensure homogeneous samples with built-in low vibration vortex.

Auto Sample
Starts sample measurement automatically once sample arm is closed.

Certainty in Real Time
Impurity and air bubble recognition with Sample Control and Blank Control

True Path Technology
Optics providing exact path lengths with two fixed anchor points. No drift over lifetime.

Flexible Unit Control
Computer (Windows & Mac)
Built-in Touchscreen
Smartphone/Tablet (Android OS & iOS)



NP80
NanoVolume and Cuvette

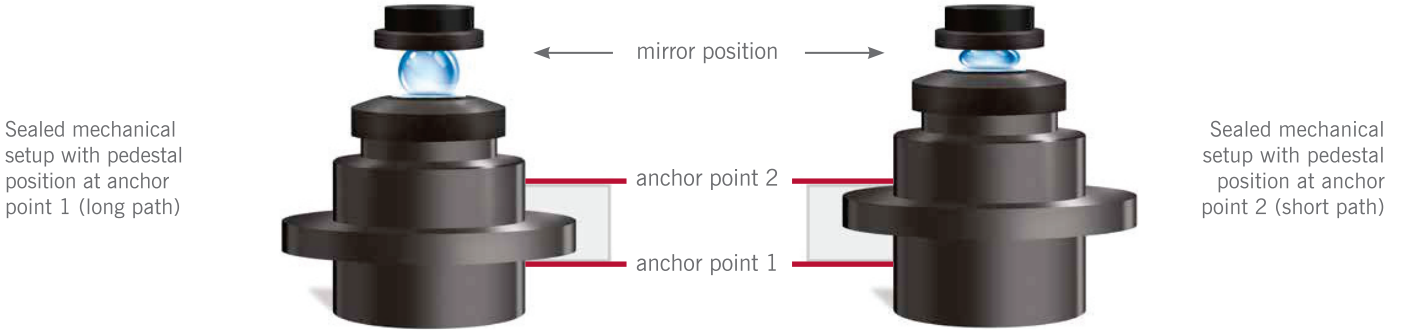


N120
12 Channel NanoVolume

True Path Technology Lifetime Accuracy Guaranteed

Precise Path Length Setting

Implens True Path Technology works with two accurate path lengths defined by fixed anchor points. To guarantee path length precision, the enclosed pedestal is engineered with high precision metal parts. Positioning of the pedestal at anchor point 1 correlates with the longer path length and anchor point 2 with the shorter path length setting (anchor points are indicated in red).



Movement between the two path lengths is controlled by a magnet. As this magnet can only position the pedestal at one of the two path lengths and cannot stop anywhere in between, path length drift is eliminated. The NanoPhotometer® is therefore recalibration-free over the lifetime of the instrument Implen guarantees lifetime accuracy.

LIMS

Data Export/Saving

Data can be saved on the internal storage (File Server), network folder or USB flash drive.
REST API for LIMS integration.
32/128 GB onboard data storage.

Data Export/Printing

Print to AirPrint and HP Universal Driver compatible printers as well as DYMO label printers.

WiFi HotSpot LAN

Endless Connectivity

Built-in File Server for data access from Windows and Mac computers.



N60/N50
NanoVolume



C40
Cuvette

Battery Powered

Up to 8 hours battery operation.

IQ OQ

IQ/OQ Documentation

Consists of a liquid and/or solid NIST traceable secondary standard depending on the configuration and specifications of the NanoPhotometer® model.

CFR 21

CFR21 Software

Optional 21 CFR Part 11 software provides password protected user accounts, data integrity, electronic signatures and audit trail functionality.

NanoPhotometer® NP80
NanoVolume & Cuvette









Detection Range	Min	Max
dsDNA	1 ng/µl	16,500 ng/µl
BSA	0.03 mg/ml	478 mg/ml

NanoPhotometer® N60/N50
NanoVolume







Model	N60		N50	
Detection Range	Min	Max	Min	Max
dsDNA	1 ng/µl	16,500 ng/µl	5 ng/µl	7,500 ng/µl
BSA	0.03 mg/ml	478 mg/ml	0.15 mg/ml	217 mg/ml

NanoPhotometer® N120

12 Channel NanoVolume

Fast High Throughput

With full scan capability range from 200 - 900 nm for rapid and complete sample analysis in as little as 1.7 seconds per sample, the NanoPhotometer® N120 record breaking design will exceed expectations. Measure 96 samples up to 100 times faster with 33% less operational steps required when compared to other scanning measurement methods such as most microplate readers. In addition to saving hands-on time with the one-step Auto Sample feature, you attain detailed information for each sample including full high resolution scan results along with sample purity ratios.

Stand-Alone Operation

Save valuable bench space. An integrated computer with an Intel Celeron Processor and 128 GB of data storage eliminates the need for a dedicated external computer. The compact, all-in-one design including a 7 glove compatible touch display and a built-in battery pack significantly reduces the bench top footprint of the device. Multi sample spectroscopy can now be taken under a laminar flow hood/clean bench, shared between labs or out into the field.

Accurate

Patented Sample Compression Technology with guaranteed fixed path lengths allows for unmatched accuracy and precision. Independent from surface tension and free from evaporation. Lifetime accuracy guaranteed.

Consistent Results

Reliably analyze up to 12 samples per run over a wide concentration range (2 to 8,000 ng/µl for dsDNA). In comparison to other technologies, the NanoPhotometer® N120 provides trustworthy data for the accurate determination of 260/280 and 260/230 ratios. Blank Control and Sample Control monitor the entire measurement process and will highlight readings that are not within the expected purity range.

Easy

NPOS Operating System: intuitive graphical user interfaces providing one-step method access, pre-programmed and customizable applications based on the ultimate level of data security. Import and export of sample IDs from Excel files is available for fast and automated sample mapping.

Performance Accelerated





Detection Range	Min	Max	Mean %CV*
dsDNA	2 ng/µl	8,000 ng/µl	0.77%; 312.9 ng/µl
BSA	0.06 mg/ml	230 mg/ml	1.01%; 5.19 mg/ml

*Based on 12 samples measured 16 times

NanoPhotometer® Model Comparison





Model	**N120** 12 Channel NanoVolume	**NP80** NanoVolume & Cuvette
Sample Number	1 - 12	1
Minimum Sample Volume	2 µl	0.3 µl
Full Spectrum Scan	200 - 900 nm	200 - 900 nm
NanoVolume Detection Range	dsDNA: 2 - 8,000 ng/µl BSA: 0.06 - 230 mg/ml	dsDNA: 1 - 16,500 ng/µl BSA: 0.03 - 478 mg/ml
Cuvette Detection Range	N/A	dsDNA: 0.1 - 130 ng/µl BSA: 0.003 - 3.7 mg /ml
Path Lengths	1 and 0.125 mm	0.67 and 0.07 mm
Built-in Cuvette	○	●
Built-in Vortex	○	●
Built-in Battery Pack	Optional	Optional
IQ/OQ Package	Optional	Optional
CFR21 Software	Optional	Optional
Recalibration-Free	●	●







N60 NanoVolume	**N50** NanoVolume	**C40** Cuvette
1	1	1
0.3 µl	0.3 µl	Depends on Cuvette
200 - 900 nm	200 - 650 nm	200 - 900 nm
dsDNA: 1 - 16,500 ng/µl BSA: 0.03 - 478 mg/ml	dsDNA: 5 - 7,500 ng/µl BSA: 0.15 - 217 mg/ml	○
N/A	N/A	dsDNA: 0.1 - 130 ng/µl BSA: 0.003 - 3.7 mg/ml
0.67 and 0.07 mm	0.67 and 0.07 mm	10, 5, 2, 1, 0.5 mm
○	○	●
●	○	○
Optional	○	
Optional	○	
Optional	○	
●	●	●

NanoPhotometer® C40

Cuvette

More Than Just a Cuvette Spectrophotometer





Detection Range	Min	Max
dsDNA	0.1 ng/µl	130 ng/µl
BSA	0.003 mg/ml	3.7 mg/ml

The OD600 DiluPhotometer™





DiluCell™

The DiluCell™ 10 allows an automatic 1/10 dilution of the sample.



Small & Portable

The DiluPhotometer™ is small and portable for fast quantification.



DiluCell™

Dilute your cells automatically.

OD600 Values

Fast and easy determination of cell density at 600 nm.

Bradford Assay

Quantify protein with Bradford protein assay.

Battery Powered

Simple to charge and take with you anywhere.

Applications

Nucleic Acids

Easy one-step analysis of dsDNA, ssDNA, RNA, miRNA and oligonucleotides including dye concentration and frequency of dye incorporation determination for conjugated nucleic acids. A full scan graph and table with raw data points including 260/280 and 260/230 purity ratios are provided.

Proteins, Peptides and Antibodies

Protein, peptide and antibody quantification using direct Protein UV/280 measurements. Adjust to the appropriate wavelength for the sample and select from preprogrammed dye labels or create custom dye methods to determine degree of labeling of dye conjugated proteins. Results are provided as OD1 value or as concentration reading based on the extinction coefficient and selected wavelength. A full scan graph and table with raw data points including 260/280 purity ratios are provided.

Protein Assays

Quantify proteins using preprogrammed colorimetric assays including BCA, Bradford and Lowry.

Kinetics

Measurement of change in absorbance as a function of time at a fixed wavelength. The change of absorbance per minute (A/min), concentration (A/min x factor) and correlation coefficient (R^2 calculated from a best fit of the data points) are displayed.

OD600 Cell Density

Absorbance readings at 600 nm. Bacterial cell cultures are routinely grown until the absorbance at 600 nm (known as OD600) reaches approximately 0.4 OD prior to induction or harvesting. Preprogrammed calculation method to determine cells/ml is available.

Wavescan

Record the absorption (Abs) or transmission (%T, cuvette mode only) spectrum in a drop or in a cuvette for a custom range between two user defined wavelengths. Automatic identification of peak height and position to determine the characteristics of any sample absorbing between 200 and 900 nm.

Standard Curve

Generate a standard curve at a single user specified wavelength between 200 and 900 nm. Prepare a multi-point calibration curve from standards of known concentration to quantify unknown samples in a drop or in a cuvette. Ability to store the curve as a method to be recalled for future experiments.

Other Available Apps

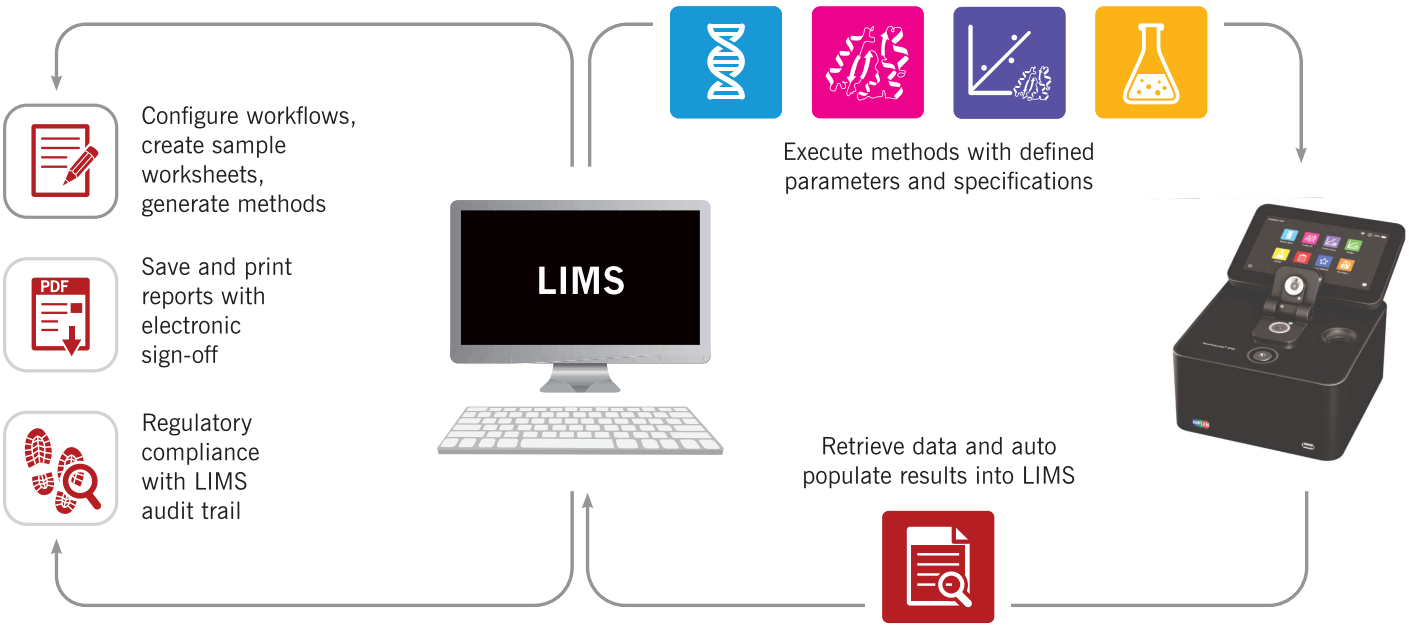
Wavelength, Concentration, Absorbance Ratio, Custom

CFR21 Software

The CFR21 software complies with FDA 21 CFR Part 11 requirements and is an optional software tool ideal for GxP laboratories, which require proper electronic record keeping. It includes user management, access control, electronic signatures, data integrity, security, and audit trail functionality.



LIMS Integration



IQ/OQ Package

International Standard

To comply with international standards in regulated environments, documented verification that your instrument is installed and functioning according to its intended use is required. Our Installation Qualification and Operation Qualification (IQ/OQ) package provides conforming data to document and verify that your instrument is working according to specifications in case of an audit. Even though the NanoPhotometer® is considered recalibration-free, also non-regulated laboratories may utilize the IQ/OQ package, since it provides added peace of mind that the system is working appropriately and producing high-quality data.

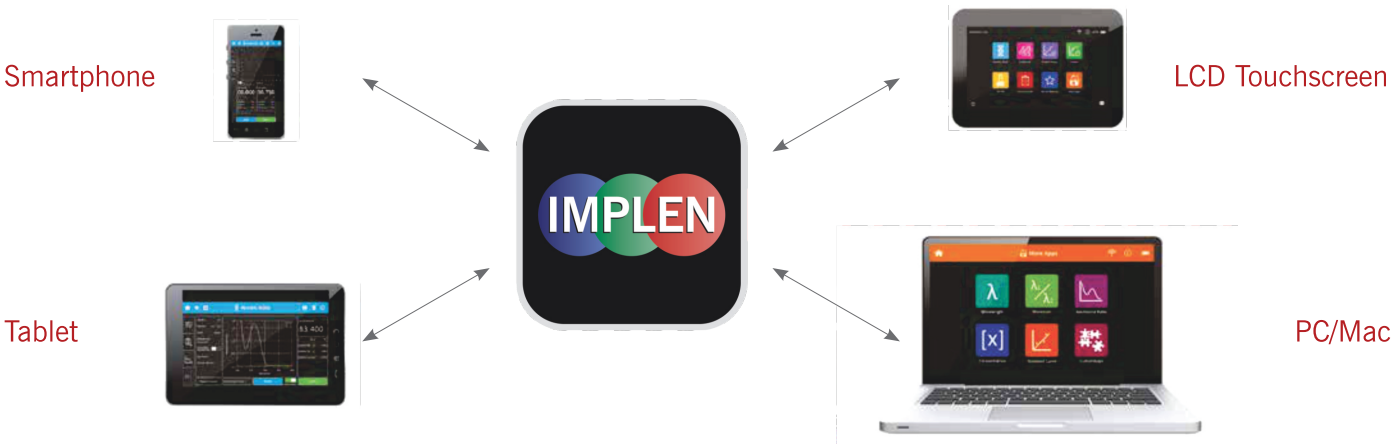




IQ/OQ Straightforward

The NanoPhotometer® IQ/OQ package consists of a liquid and/or a solid NIST traceable secondary standard and a corresponding Excel based software tool depending on the type of instrument. The liquid standard is based on a non-toxic compound absorbing at 280 nm to cover the area of most interest for Life Science applications. All secondary standards come with a certificate of compliance. The Excel based software produces an automated report displaying all relevant data for an audit.

NPOS Operating System



The most powerful operating system in UV/Vis spectrophotometry provides flexibility and mobility with its web based interface running stand-alone utilizing the built-in 7 inch screen or through access from multiple devices (computer, tablet and smartphone*) and systems (Windows, Mac, Android OS, and iOS). Pre-programmed and customizable applications are available with intuitive interfaces and a one-step method access for easy data analysis and data/method saving. The Linux based NPOS (NanoPhotometer® Operating System) also provides the ultimate level of data security.

*Smartphone is not available with the N120

Endless Connectivity



Every NanoPhotometer® is equipped with WiFi, built-in HotSpot, LAN, HDMI and USB A/B interface connections for remote instrument control, data accessibility and printing. NPOS supports AirPrint compatible printers, printers compatible with the HP Universal driver, and DYMO label printers (cryo and standard labels).

Technical Specifications

NanoVolume Performance	
Detection Range dsDNA	N60, NP80: 1 - 16,500 ng/µl N50: 5 - 7,500 ng/µl N120: 2 - 8,000 ng/µl
Detection Range BSA	N60, NP80: 0.03 - 478 mg/ml N50: 0.15 - 217 mg/ml N120: 0.06 - 230 mg/ml
Sample Volume	N50, N60, NP80: 0.3 - 2 µl N120: 2 - 3.5 µl
Photometric Range (10 mm equivalent)	N60, NP80: 0.02 - 330 A N50: 0.1 - 150 A N120: 0.04 - 160 A
Path Length	N50, N60, NP80: 0.67 & 0.07 mm N120: 1 and 0.125 mm
Dilution Factor	N50, N60, NP80: 15 and 140 N120: 10 and 80
Vortex	N60, NP80: 2,800 rpm Tube size up to 2.0 ml

Cuvette Performance – NP80 & C40	
Detection Range dsDNA	0.1 - 130 ng/µl
Detection Range BSA	0.003 - 3.7 mg/ml
Photometric Range	0 - 2.6 A
Center Height (Z-Height)	8.5 mm
Cell Types	Outside dimension 12.5 x 12.5 mm
Heating	37 °C ± 0.5 °C

Processing Power & Compatibility	
Operating System	Linux based NPOS
Onboard Processor	Intel Celeron dual core 2.4 GHz
Internal Data Storage	C40, N50, N60, NP80: 32 GB N120: 128 GB
In & Output Ports	2x USB A, USB B, HDMI, Ethernet, WiFi
Software Compatibility	Windows 8, 10 (32 & 64 bit) OS X (Intel x86 and Apple M1) iOS and Android OS

Optical Specifications	
Wavelength Scan Range	C40, N60, NP80, N120: 200 - 900 nm N50: 200 - 650 nm
Measure Time For Full Scan Range	C40, N50, N60, NP80: 2.5 - 4.0 sec N120: 1.7 - 2.5 sec per sample
Wavelength Reproducibility	C40, N60, NP80, N120: ± 0.2 nm N50: ± 1 nm
Wavelength Accuracy	C40, N60, NP80, N120: ± 0.75 nm N50: ± 1.5 nm
Bandwidth	C40, N60, NP80: < 1.5 nm N50: < 3 nm N120: < 2.5 nm
Absorbance Reproducibility	C40, NP80 (Cuvette): < 0.002 A @ 0 - 0.3 A @ 280 nm CV < 1% @ 0.3 - 2.0 A @ 280 nm N50 (Lid 15): < 0.004 A @ 0 - 0.3 A @ 280 nm CV < 1% @ 0.3 - 1.5 A @ 280 nm N60, NP80 (Lid 15): < 0.002 A @ 0 - 0.3 A @ 280 nm CV < 1% @ 0.3 - 1.7 A @ 280 nm N120 (Lid 10): < 0.004 A @ 0 - 0.3 A @ 280 nm CV < 0.4% @ 0.8 A @ 280 nm
Absorbance Accuracy	< 1.75% @ 0.7 A @ 280 nm of the reading
Stray Light	N60, NP80, C40: < 0.5% @ 240 nm using NaI N50: < 2% @ 240 nm using NaI N120: < 1% @ 240 nm using NaI
Optical Arrangement	C40, N50, N60, NP80: 1 x 4096 CMOS Array N120: 1 x 3648 CCD Array
Lamp | Lifetime	Xenon flash lamp | 10^9 flashes, up to 10 years

General Specifications	
Main Body Size	200 x 200 x 120 mm
Weight	3.8 - 5.2 kg depending on configuration
Operating Voltage	90 - 250 V ± 10%, 50/60 Hz, 90 W, 18/19 VDC
Display	1024 x 600 pixels; glove compatible touchscreen
Built-in Battery Pack: Optional rechargeable lithium ion battery	C40, N60, NP80: 95 Wh, 6.6 Ah, 8 h N120: 47.5 Wh, 3.3 Ah, 3 h Min. charging cycles: 800
Certification	CE, IEC 61010-1:2012 and EN 61326-1:2013
Battery Certification	IEC 62133 and UN38.3 transport test
Security	Slot for Kensington lock

Features and specifications are subject to change without notice.

ISO 9001:2015 CERTIFIED | MSME सूक्ष्म, लघु एवं मध्यम उद्यम MICRO, SMALL & MEDIUM ENTERPRISES

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Your Knowledge... Our Solution...!

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